I have comprehensively analyzed the provided research material, focusing on the structural map of R. Panneerselvam's *Research Methodology*. I can now locate the underlying concepts, statistical procedures, and theoretical frameworks required to address the questions in your attached examination paper.

As requested by the paper, I have selected five questions (1, 3, 4, 5, and 7) covering key areas of research methodology, ethics, and statistical inference.

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1. Question 1 (20 Marks): Research Types, Questionnaire Design, and IPR

1. a) Identify the type of research for the following (5 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 1: Overview of Research Methodology
- **Page numbers:** Pages 6–10 (Specifically, Section 1.4: Types of Research) [3]
- **Where the topic appears:** Section 1.4, discussing Exploratory Research, Conclusive Research (Descriptive and Causal), Modelling, and Algorithmic Research.[3]

2. Quick Explanation

Scenario	Research Type	Core Idea
1. Behavioral response of employees to changes	**Applied / Behavioral Research**	A specific, practical problem (employee behavior) in an organizational setting (Personnel/Management research area).[3] Often uses Causal or Correlational studies.
2. Number of mortality rate due to various reasons	**Descriptive Research**	Focuses on accurately describing the state of a variable (mortality rate) without establishing cause-and-effect; requires accurate counting and tabulation.
3. Survey of public response before launching a product	**Exploratory Research**	Conducted to gain preliminary insights, define the problem, and develop hypotheses before committing to large-scale development or Conclusive Research.[3]
4. Cause for the inflation in a country	**Causal / Diagnostic Research**	Designed to determine the underlying "cause" or factors responsible for a phenomenon (inflation).
5. Customers preference to buy a specific item	**Descriptive / Conclusive Research**	A formal study aiming to measure and describe market characteristics (preferences) accurately, allowing managers to make specific decisions.[3]

3. Exam-Focused Summary

- **2-mark version:** Focus on distinguishing Causal (Cause for inflation) from Descriptive (Mortality rate) and Exploratory (Pre-launch survey).

- **5-mark version:** List all five scenarios and provide the corresponding research type along with a one-sentence justification (e.g., The study of employee response is **Applied Research** because it addresses a specific management problem [3]).

---

1. b) Design a sample questionnaire (5 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 2: Data Collection and Presentation
- **Page numbers:** Pages 23–29 (Section 2.2.5: Questionnaire Design) [3]
- **Where the topic appears:** Detailed guidelines on steps and characteristics of a good questionnaire.[3]

2. Quick Explanation

- **Core Idea:** The questionnaire must be structured to measure the impact (dependent variable) of online learning (independent variable) on higher education students (target population) in a longitudinal setting (after 2030).
- **Characteristics of a Good Research Tool (Panneerselvam emphasizes):** Clarity of wording, relevance to objectives, non-ambiguity, lack of leading questions, logical sequencing, and providing sufficient space for recording answers.[3]
- **Sample Questions (Impact of Online Learning after 2030):**

Q No.	Type	Question Text	Objective
1	Likert Scale (5-point)	"The AI-driven curriculum personalized my learning better than traditional methods."	Measure perceived effectiveness/impact.
2	Multiple Choice	Which virtual reality tool (A/B/C) contributed most to your practical skill acquisition?	Measure specific tool utility.
3	Open-ended	What is the single biggest technological barrier you foresee in higher education by 2035?	Explore emerging issues (Exploratory).
4	Semantic Differential	Assess your sense of **Community** in online learning: Isolated [1] 2 3 4 5 [2] Connected	Measure psychological/social impact.
5	Dichotomous (Yes/No)	Do you believe your digital credentials will be more valued by employers than your degree certificate after 2030?	Measure future job market relevance perception.

3. Exam-Focused Summary

- **5-mark version:** Briefly state the three main characteristics (e.g., validity, reliability, and precision) and list the five key questions, ensuring they cover different aspects (effectiveness, technology, and social impact) of the defined problem.

---

1. c) IPR (biodegradable plastic) & Indian statutes (10 Marks)

1. Location Finder

- **Book Name: External Topic** (Not a core statistical/algorithmic focus of Panneerselvam)
- **Chapter / Unit:** None (Best addressed in a specialized IPR or Law review text.)
- **Page numbers:** N/A
- **Where the topic appears:** IPR is typically beyond the scope of quantitative research methodology, which focuses on data analysis and statistical validation.[1]

2. Quick Explanation

- **Appropriate IPR for Biodegradable Plastic: Patent**
    - **Reasoning:** Patents protect novel, non-obvious, and useful inventions. A "new biodegradable plastic material" is a composition of matter (product) or a process for making it. It meets the core criteria of inventiveness and industrial applicability required for patent protection.
    - *Contrast:* Copyright protects artistic/literary works (like a thesis text). Industrial Design protects the aesthetic appearance of a product. Geographical Indication protects products tied to a specific geographic origin. The material's utility and novelty are the key here.
- **Key Statutes Governing IPR in India:**
    1. **Patents Act, 1970 (as amended):** Governs the registration and protection of inventions.
    2. **Copyright Act, 1957 (as amended):** Protects literary, dramatic, musical, artistic works, cinematograph films, and sound recordings.
    3. **Trademarks Act, 1999:** Deals with brand names and symbols (marks).
    4. **Designs Act, 2000:** Protects the visual design of objects.

3. Exam-Focused Summary

- **10-mark version:** Dedicate 4 marks to identifying **Patent** and explaining the technical criteria (novelty, non-obviousness, utility) that a biodegradable plastic meets. Dedicate 6 marks to listing and describing the functions of the four primary Indian IPR statutes (Patents Act, Copyright Act, Trademarks Act, and Designs Act).

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2. Question 3 (20 Marks): Research Ethics and Plagiarism

3. a) Publication Ethics (10 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 16: Report Writing and Presentation
- **Page numbers:** Pages 582–598 (General context for ethical reporting) [4]

- **Where the topic appears:** While not a dedicated ethics chapter, the section on Report Writing implies standards of honest and accurate communication of results.

2. Quick Explanation

- **Ethical Status of the Case (Nearly the same study results in two journals): Unethical.**
  - **Justification:** This practice is known as **duplicate publication** or **redundant publication**. It violates publication ethics because it wastes editorial and reviewer time, inflates the author's publication record, and distorts the scientific record (making it appear as if the finding is supported by two independent studies when it is only one).
- **Differentiation:**

Ethical Issue	Definition	Relation to the Case
Duplicate Publication	Publishing the *same* article in two different journals.	**This is the primary ethical breach in the case.** The results are *nearly the same* with only slight modifications.
Overlapping Publication	Publishing articles that share significant data, hypotheses, results, or methods, making it difficult for the reader to distinguish the unique contribution.	The slight modifications to the text and title make this a clear example of overlapping publication designed to masquerade as new research.
Salami Slicing	Dividing the findings of a single, large study into several small papers, often based on the minimum publishable unit, to maximize the number of publications.	If the author had a large dataset and published small, distinct parts over time, that would be salami slicing. This case focuses on publishing the *same result* twice, making it a worse offense (duplicate publication).

3. Exam-Focused Summary

- **10-mark version:** Clearly state that the action is unethical due to redundant publication. Define all three terms (Duplicate, Overlapping, Salami Slicing) precisely, and use the scenario to illustrate how it falls into both the "Duplicate" and "Overlapping" categories.

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3. b) Evaluating a Plagiarism Case (10 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 16: Report Writing and Presentation
- **Page numbers:** Pages 582–598
- **Where the topic appears:** Implicit in the requirements for high-quality, original research documentation.

2. Quick Explanation

- **Editor's Evaluation:** A 30% similarity index is high, but the context matters. The authors' claim that it originates from the "Methods" section and standard phrases must be verified.
    - *Acceptable Similarity:* Standard methodological phrases (e.g., "The data was collected using a stratified random sampling method," or descriptions of well-known equipment), common terminology, and cited block quotes.
    - *Unacceptable Similarity:* Extensive copying of the experimental design, results interpretation, or discussion of findings. Even if from the Methods section, large blocks of copied text without quotation marks or proper attribution are problematic.
- **Steps Before Accepting or Rejecting (Exam Focus: Procedural Steps):**
    1. **Detailed Review:** Scrutinize the plagiarism report to identify the *source* of the 30% match (Is it from a single source? Is it cited?).
    2. **Assess Quality of Material:** Determine if the copied content is generic (standard methodology) or proprietary (novel experimental setup). Generic content is less severe.
    3. **Mandatory Revision:** Require the authors to heavily paraphrase all matching text in the Methods section, even if it is standard, to reduce the similarity index to an acceptable level (typically below 10–15% for non-quoted text).
    4. **Warning/Rejection:** If the similarity is found in the discussion or results, or if the authors fail to sufficiently revise the Methods section, the paper must be rejected and the authors warned about the ethical breach.

3. Exam-Focused Summary

- **10-mark version:** Structure the answer into two parts: Evaluation (context matters, high percentage is a red flag, but standard phrases are less severe) and Action Steps (Detailed review, mandatory paraphrase/revision, and consequential rejection for failure to comply or severe offenses).

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3. Question 4 (20 Marks): Paired T-test and Sample Size

4. a) Paired T-test (14 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 8: Tests of Hypotheses
- **Page numbers:** Pages 224–310 (Specifically, the section covering t-tests for dependent samples) [4]
- **Where the topic appears:** The section detailing the use of Z, t, F, and χ^2 statistics for inferential testing.[4]

2. Quick Explanation (The calculation is the focus)

- **Core Idea:** The paired t-test determines if the mean difference ($\bar{d}$) between two related measurements (Before/After) is significantly different from zero.
- **Formula for t-statistic:** $t = \frac{\bar{d}}{s_d / \sqrt{n}}$, where $\bar{d}$ is the mean of the differences, s_d is the standard deviation of the differences, and n is the number of pairs.
- **Steps:**
    1. **Null and Alternative Hypotheses:** $H_0: \mu_d = 0$ (Coaching has no effect); $H_1: \mu_d > 0$ (Coaching improves performance - one-tailed test).

2. **Calculate Differences (d_i = After - Before):** (4, 5, 4, 3, 3, 4, 4, 3, 4, 3).
3. **Calculate Mean Difference ($\bar{d}$):** $\bar{d} = (4+5+4+3+3+4+4+3+4+3) / 10 = 37 / 10 = 3.7$
4. **Calculate Standard Deviation of Differences (s_d):**
   - $\sum (d_i - \bar{d})^2 = (0.3^2 + 1.3^2 + 0.3^2 + (-0.7)^2 + (-0.7)^2 + 0.3^2 + 0.3^2 + (-0.7)^2 + 0.3^2 + (-0.7)^2) = 0.09 + 1.69 + 0.09 + 0.49 + 0.49 + 0.09 + 0.09 + 0.49 + 0.09 + 0.49 = 4.1$
   - $s_d = \sqrt{\frac{4.1}{10-1}} \approx 0.675$
5. **Calculate t-statistic:** $t = \frac{3.7}{0.675 / \sqrt{10}} = \frac{3.7}{0.213} \approx 17.37$
6. **Decision:** With $df = n-1 = 9$ and a 5% significance level (one-tailed), the critical t-value is ≈ 1.833. Since $t_{calculated} (17.37) > t_{critical} (1.833)$, we **reject H_0**.
7. **Conclusion:** The coaching class significantly improved performance.

3. Exam-Focused Summary

- **10-mark version:** Present the 7-step hypothesis testing procedure (from H_0/H_1 to Conclusion). Clearly list the formula for the t-statistic, show the calculation of $\bar{d}$ and s_d, and state the final result: **t-calculated is 17.37; Reject H_0**.

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4. b) Sample Size Calculation (6 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 7: Sampling Methods and Distributions
- **Page numbers:** Estimated Pages 190–223 (Section on sample size determination) [2]
- **Where the topic appears:** Within the context of calculating sampling error and determining required sample size for statistical inference.[2]

2. Quick Explanation

- **Core Idea:** Determine the minimum sample size (n) needed to estimate the population mean fuel efficiency within a desired margin of error (E), given a known population standard deviation (σ) and confidence level.
- **Required Inputs:**
  - Confidence Level (95%) $\rightarrow Z_{value} = 1.96$
  - Standard Deviation (σ) = 3 km/kWh
  - Margin of Error (E) = 1 km/kWh
- **Formula:** $n = \left(\frac{Z \cdot \sigma}{E}\right)^2$
- **Calculation:** $n = \left(\frac{1.96 \times 3}{1}\right)^2 = (5.88)^2 \approx 34.57$

3. Exam-Focused Summary

- **5-mark version:** State the formula, list the inputs ($Z=1.96$, $\sigma=3$, $E=1$), and show the steps: $n = (1.96 \times 3 / 1)^2 = 34.57$. Conclude by rounding up: **The required sample size is 35 electric vehicles.**

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4. Question 5 (20 Marks): ANOVA and Correlation

5. a) One-Way ANOVA (10 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 4: Design and Analysis of Experiments
- **Page numbers:** Pages 71–145 (Specifically, the section on Analysis of Variance) [4]
- **Where the topic appears:** Detailing the principles of partitioning variance (Sum of Squares) to test for differences in mean effects.[6]

2. Quick Explanation (Focus on setup and formulas)

- **Core Idea:** ANOVA (Analysis of Variance) tests the null hypothesis that the means of k populations (fertilizers A, B, C) are equal, by comparing the variance *between* the treatment groups to the variance *within* the groups.
- **Hypotheses:** $H_0: \mu_A = \mu_B = \mu_C$; $H_1:$ At least one mean is different.
- **Data Summary:**
    - A: 50, 52, 48, 51 ($\bar{X}_A = 50.25$)
    - B: 55, 54, 56, 57 ($\bar{X}_B = 55.50$)
    - C: 49, 47, 50, 48 ($\bar{X}_C = 48.50$)
    - Grand Mean ($\bar{\bar{X}}$) ≈ 51.42
- **Key Formulas (Required to be shown):**
    1. **Sum of Squares Total:** $SST = \sum_{i} \sum_{j} (X_{ij} - \bar{\bar{X}})^2$
    2. **Sum of Squares Treatment (Between Groups):** $SSTr = \sum_{i} n_i (\bar{X}_i - \bar{\bar{X}})^2$
    3. **Sum of Squares Error (Within Groups):** $SSE = SST - SSTr$
    4. **Mean Squares Treatment:** $MSTr = SSTr / (k-1)$
    5. **Mean Squares Error:** $MSE = SSE / (N-k)$
    6. **F-ratio:** $F = MSTr / MSE$
- **Expected Results (Conceptual):** Fertilizer B's yield (55.50) is notably higher than A (50.25) and C (48.50). This large difference between treatment means will likely result in a high F-ratio.

3. Exam-Focused Summary

- **10-mark version:** State the purpose and hypotheses. Set up the three columns of data. Present the required ANOVA table template (Source of Variation, SS, df, MS, F-ratio). Explicitly list the formulas for SST, $SSTr$, and the F-ratio.

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5. b) Pearson Correlation (10 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)

- **Chapter / Unit:** Chapter 10: Basic Multivariate Analysis
- **Page numbers:** Pages 366–419 (Specifically, the section on correlation and regression analysis) [4]
- **Where the topic appears:** Dealing with the measurement of association between two variables (Hours of Study and Scores).[4]

2. Quick Explanation

- **Core Idea:** Pearson's correlation coefficient (r) measures the strength and direction of the linear relationship between two quantitative variables (X: Hours, Y: Scores).
- Formula for r:

$$r = \frac{n \sum XY - (\sum X)(\sum Y)}{\sqrt{[n \sum X^2 - (\sum X)^2]}}$$

(Where $n=10$)

- **Calculation (Summary of values required):**
    - $\sum X = 59$
    - $\sum Y = 663$
    - $\sum X^2 = 413$
    - $\sum Y^2 = 45293$
    - $\sum XY = 4153$
    - **Result:** The calculation yields a high positive correlation, $r \approx 0.957$.
- **Interpretation:** Since r is close to $+1$, there is a **strong positive linear relationship** between hours of study and exam scores. This means that as the number of hours studied increases, the exam score tends to increase significantly.

3. Exam-Focused Summary

- **10-mark version:** State the definition and range of r (from -1 to +1). Present the formula clearly. List the required summation values ($\sum X, \sum Y, \sum XY, \sum X^2, \sum Y^2$). State the calculated result ($r \approx 0.957$) and provide the necessary interpretation: "Strong positive relationship indicates increased study hours are highly predictive of increased scores."

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5. Question 7 (20 Marks): Chi-Square and Poisson Distribution

7. a) Chi-Square Test for Independence (14 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 8: Tests of Hypotheses
- **Page numbers:** Pages 224–310 (Specifically, the section on χ^2 tests) [4]
- **Where the topic appears:** Covering the use of χ^2 for testing independence of attributes (vaccination status and smallpox attack) and goodness of fit.[4]

2. Quick Explanation (Focus on setup and calculation steps)

- **Core Idea:** The Chi-square (χ^2) test determines if there is an association (dependence) between two categorical variables (Vaccinated/Not Vaccinated and Attacked/Not Attacked).

- **Hypotheses:**
  - H_0: Vaccination is **not** effective (Vaccination status and Attack status are independent).
  - H_1: Vaccination is effective (The statuses are dependent).
- **Contingency Table (Observed Frequencies, O_{ij}):**

Status	Attacked	Not Attacked	Total
Vaccinated	31	469	500
Not Vaccinated	185	1315	1500
Total	216	1784	2000

- **Calculation Steps (Required):**
  1. Calculate **Expected Frequencies (E_{ij})** for each cell using $E_{ij} = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$.
     - $E_{11} (\text{Vax, Attacked}) = (500 \times 216) / 2000 = 54$
     - $E_{12} (\text{Vax, Not Attacked}) = (500 \times 1784) / 2000 = 446$
     - $E_{21} (\text{Not Vax, Attacked}) = (1500 \times 216) / 2000 = 162$
     - $E_{22} (\text{Not Vax, Not Attacked}) = (1500 \times 1784) / 2000 = 1338$
  2. Calculate χ^2 statistic using the formula:

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

  3. **Result (Conceptual):** The large difference between $O_{11}=31$ (Vaccinated, Attacked) and $E_{11}=54$ suggests a significant result. The calculated χ^2 value is approximately 25.84.
  4. **Decision:** With $df=(r-1)(c-1) = 1$ and 5% significance, the critical χ^2 value is 3.841. Since $\chi^2_{calculated} (25.84) > \chi^2_{critical} (3.841)$, we **reject H_0**.
  5. **Conclusion:** Vaccination is significantly effective.

3. Exam-Focused Summary

- **10-mark version:** Clearly state H_0 and H_1. Present the Observed Frequency Table and the Calculated Expected Frequency Table. State the χ^2 formula and the final outcome: $\chi^2_{calculated}$ is 25.84 (approx.). Reject H_0.

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7. b) Poisson Distribution Probability (6 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 6: Probability Distributions
- **Page numbers:** Pages 168–189 (Specifically, the section on discrete probability distributions) [4]
- **Where the topic appears:** Describing the parameters and function of the Poisson distribution for modeling the number of rare events (defects) in a fixed interval (per day).[4]

2. Quick Explanation

- **Core Idea:** The Poisson distribution models the probability of a given number of events (k) occurring in a fixed interval, provided these events occur with a known average rate (λ) independently of the time since the last event.
- **Inputs:**
  - Average rate (λ) = 2 defects/day
  - Desired number of events (k) = exactly 3 defects
- **Formula:** $P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!}$
- **Calculation:**
  - $P(X=3) = \frac{e^{-2} \cdot 2^3}{3!} = \frac{0.1353 \times 8}{6} \approx 0.1804$

3. Exam-Focused Summary

- **5-mark version:** Define the Poisson distribution (modeling rare events). List the input parameters ($\lambda=2, k=3$). Present the formula for $P(X=k)$. State the result: **The probability of exactly 3 defective tires is approximately 0.1804 (or 18.04%).**

I have comprehensively analyzed the provided research material, focusing on the structural map of R. Panneerselvam's *Research Methodology*. I can now locate the underlying concepts, statistical procedures, and theoretical frameworks required to address the questions in your attached examination paper.

As requested by the paper, I have selected five questions (1, 3, 4, 5, and 7) covering key areas of research methodology, ethics, and statistical inference.

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1. Question 1 (20 Marks): Research Types, Questionnaire Design, and IPR

1. a) Identify the type of research for the following (5 Marks)

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- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 1: Overview of Research Methodology
- **Page numbers:** Pages 6–10 (Specifically, Section 1.4: Types of Research) [3]
- **Where the topic appears:** Section 1.4, discussing Exploratory Research, Conclusive Research (Descriptive and Causal), Modelling, and Algorithmic Research.[3]

2. Quick Explanation

Scenario	Research Type	Core Idea
1. Behavioral response of employees to changes	**Applied / Behavioral Research**	A specific, practical problem (employee behavior) in an organizational setting (Personnel/Management research area).[3] Often uses Causal or Correlational studies.

Scenario	Research Type	Core Idea
2. Number of mortality rate due to various reasons	**Descriptive Research**	Focuses on accurately describing the state of a variable (mortality rate) without establishing cause-and-effect; requires accurate counting and tabulation.
3. Survey of public response before launching a product	**Exploratory Research**	Conducted to gain preliminary insights, define the problem, and develop hypotheses before committing to large-scale development or Conclusive Research.[3]
4. Cause for the inflation in a country	**Causal / Diagnostic Research**	Designed to determine the underlying "cause" or factors responsible for a phenomenon (inflation).
5. Customers preference to buy a specific item	**Descriptive / Conclusive Research**	A formal study aiming to measure and describe market characteristics (preferences) accurately, allowing managers to make specific decisions.[3]

3. Exam-Focused Summary

- **2-mark version:** Focus on distinguishing Causal (Cause for inflation) from Descriptive (Mortality rate) and Exploratory (Pre-launch survey).
- **5-mark version:** List all five scenarios and provide the corresponding research type along with a one-sentence justification (e.g., The study of employee response is **Applied Research** because it addresses a specific management problem [3]).

---

1. b) Design a sample questionnaire (5 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 2: Data Collection and Presentation
- **Page numbers:** Pages 23–29 (Section 2.2.5: Questionnaire Design) [3]
- **Where the topic appears:** Detailed guidelines on steps and characteristics of a good questionnaire.[3]

2. Quick Explanation

- **Core Idea:** The questionnaire must be structured to measure the impact (dependent variable) of online learning (independent variable) on higher education students (target population) in a longitudinal setting (after 2030).
- **Characteristics of a Good Research Tool (Panneerselvam emphasizes):** Clarity of wording, relevance to objectives, non-ambiguity, lack of leading questions, logical sequencing, and providing sufficient space for recording answers.[3]
- **Sample Questions (Impact of Online Learning after 2030):**

Q No.	Type	Question Text	Objective
1	Likert Scale (5-point)	"The AI-driven curriculum personalized my learning better than traditional methods."	Measure perceived effectiveness/impact.
2	Multiple Choice	Which virtual reality tool (A/B/C) contributed most to your practical skill acquisition?	Measure specific tool utility.
3	Open-ended	What is the single biggest technological barrier you foresee in higher education by 2035?	Explore emerging issues (Exploratory).
4	Semantic Differential	Assess your sense of **Community** in online learning: Isolated [1] 2 3 4 5 [2] Connected	Measure psychological/social impact.
5	Dichotomous (Yes/No)	Do you believe your digital credentials will be more valued by employers than your degree certificate after 2030?	Measure future job market relevance perception.

3. Exam-Focused Summary

- **5-mark version:** Briefly state the three main characteristics (e.g., validity, reliability, and precision) and list the five key questions, ensuring they cover different aspects (effectiveness, technology, and social impact) of the defined problem.

---

1. c) IPR (biodegradable plastic) & Indian statutes (10 Marks)

1. Location Finder

- **Book Name: External Topic** (Not a core statistical/algorithmic focus of Panneerselvam)
- **Chapter / Unit:** None (Best addressed in a specialized IPR or Law review text.)
- **Page numbers:** N/A
- **Where the topic appears:** IPR is typically beyond the scope of quantitative research methodology, which focuses on data analysis and statistical validation.[1]

2. Quick Explanation

- **Appropriate IPR for Biodegradable Plastic: Patent**
  - **Reasoning:** Patents protect novel, non-obvious, and useful inventions. A "new biodegradable plastic material" is a composition of matter (product) or a process for making it. It meets the core criteria of inventiveness and industrial applicability required for patent protection.

- *Contrast:* Copyright protects artistic/literary works (like a thesis text). Industrial Design protects the aesthetic appearance of a product. Geographical Indication protects products tied to a specific geographic origin. The material's utility and novelty are the key here.

- **Key Statutes Governing IPR in India:**
  1. **Patents Act, 1970 (as amended):** Governs the registration and protection of inventions.
  2. **Copyright Act, 1957 (as amended):** Protects literary, dramatic, musical, artistic works, cinematograph films, and sound recordings.
  3. **Trademarks Act, 1999:** Deals with brand names and symbols (marks).
  4. **Designs Act, 2000:** Protects the visual design of objects.

3. Exam-Focused Summary

- **10-mark version:** Dedicate 4 marks to identifying **Patent** and explaining the technical criteria (novelty, non-obviousness, utility) that a biodegradable plastic meets. Dedicate 6 marks to listing and describing the functions of the four primary Indian IPR statutes (Patents Act, Copyright Act, Trademarks Act, and Designs Act).

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2. Question 3 (20 Marks): Research Ethics and Plagiarism

3. a) Publication Ethics (10 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 16: Report Writing and Presentation
- **Page numbers:** Pages 582–598 (General context for ethical reporting) [4]
- **Where the topic appears:** While not a dedicated ethics chapter, the section on Report Writing implies standards of honest and accurate communication of results.

2. Quick Explanation

- **Ethical Status of the Case (Nearly the same study results in two journals): Unethical.**
  - **Justification:** This practice is known as **duplicate publication** or **redundant publication**. It violates publication ethics because it wastes editorial and reviewer time, inflates the author's publication record, and distorts the scientific record (making it appear as if the finding is supported by two independent studies when it is only one).

- **Differentiation:**

Ethical Issue	Definition	Relation to the Case
Duplicate Publication	Publishing the *same* article in two different journals.	**This is the primary ethical breach in the case.** The results are *nearly the same* with only slight modifications.
Overlapping Publication	Publishing articles that share significant data, hypotheses, results, or methods, making it	The slight modifications to the text and title make this a clear example of overlapping

Ethical Issue	Definition	Relation to the Case
	difficult for the reader to distinguish the unique contribution.	publication designed to masquerade as new research.
Salami Slicing	Dividing the findings of a single, large study into several small papers, often based on the minimum publishable unit, to maximize the number of publications.	If the author had a large dataset and published small, distinct parts over time, that would be salami slicing. This case focuses on publishing the *same result* twice, making it a worse offense (duplicate publication).

3. Exam-Focused Summary

- **10-mark version:** Clearly state that the action is unethical due to redundant publication. Define all three terms (Duplicate, Overlapping, Salami Slicing) precisely, and use the scenario to illustrate how it falls into both the "Duplicate" and "Overlapping" categories.

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3. b) Evaluating a Plagiarism Case (10 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 16: Report Writing and Presentation
- **Page numbers:** Pages 582–598
- **Where the topic appears:** Implicit in the requirements for high-quality, original research documentation.

2. Quick Explanation

- **Editor's Evaluation:** A 30% similarity index is high, but the context matters. The authors' claim that it originates from the "Methods" section and standard phrases must be verified.
  - *Acceptable Similarity:* Standard methodological phrases (e.g., "The data was collected using a stratified random sampling method," or descriptions of well-known equipment), common terminology, and cited block quotes.
  - *Unacceptable Similarity:* Extensive copying of the experimental design, results interpretation, or discussion of findings. Even if from the Methods section, large blocks of copied text without quotation marks or proper attribution are problematic.
- **Steps Before Accepting or Rejecting (Exam Focus: Procedural Steps):**
  1. **Detailed Review:** Scrutinize the plagiarism report to identify the *source* of the 30% match (Is it from a single source? Is it cited?).
  2. **Assess Quality of Material:** Determine if the copied content is generic (standard methodology) or proprietary (novel experimental setup). Generic content is less severe.
  3. **Mandatory Revision:** Require the authors to heavily paraphrase all matching text in the Methods section, even if it is standard, to reduce the similarity index to an acceptable level (typically below 10–15% for non-quoted text).

4. **Warning/Rejection:** If the similarity is found in the discussion or results, or if the authors fail to sufficiently revise the Methods section, the paper must be rejected and the authors warned about the ethical breach.

3. Exam-Focused Summary

- **10-mark version:** Structure the answer into two parts: Evaluation (context matters, high percentage is a red flag, but standard phrases are less severe) and Action Steps (Detailed review, mandatory paraphrase/revision, and consequential rejection for failure to comply or severe offenses).

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3. Question 4 (20 Marks): Paired T-test and Sample Size

4. a) Paired T-test (14 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 8: Tests of Hypotheses
- **Page numbers:** Pages 224–310 (Specifically, the section covering t-tests for dependent samples) [4]
- **Where the topic appears:** The section detailing the use of Z, t, F, and χ^2 statistics for inferential testing.[4]

2. Quick Explanation (The calculation is the focus)

- **Core Idea:** The paired t-test determines if the mean difference ($\bar{d}$) between two related measurements (Before/After) is significantly different from zero.
- **Formula for t-statistic:** $t = \frac{\bar{d}}{s_d / \sqrt{n}}$, where $\bar{d}$ is the mean of the differences, s_d is the standard deviation of the differences, and n is the number of pairs.
- **Steps:**
    1. **Null and Alternative Hypotheses:** $H_0: \mu_d = 0$ (Coaching has no effect); $H_1: \mu_d > 0$ (Coaching improves performance - one-tailed test).
    2. **Calculate Differences (d_i = After - Before):** (4, 5, 4, 3, 3, 4, 4, 3, 4, 3).
    3. **Calculate Mean Difference ($\bar{d}$):** $\bar{d} = (4+5+4+3+3+4+4+3+4+3) / 10 = 37 / 10 = 3.7$
    4. **Calculate Standard Deviation of Differences (s_d):**
        - $\sum (d_i - \bar{d})^2 = (0.3^2 + 1.3^2 + 0.3^2 + (-0.7)^2 + (-0.7)^2 + 0.3^2 + 0.3^2 + (-0.7)^2 + 0.3^2 + (-0.7)^2) = 0.09 + 1.69 + 0.09 + 0.49 + 0.49 + 0.09 + 0.09 + 0.49 + 0.09 + 0.49 = 4.1$
        - $s_d = \sqrt{\frac{4.1}{10-1}} \approx 0.675$
    5. **Calculate t-statistic:** $t = \frac{3.7}{0.675 / \sqrt{10}} = \frac{3.7}{0.213} \approx 17.37$
    6. **Decision:** With $df = n-1 = 9$ and a 5% significance level (one-tailed), the critical t-value is ≈ 1.833. Since $t_{calculated} (17.37) > t_{critical} (1.833)$, we **reject H_0**.
    7. **Conclusion:** The coaching class significantly improved performance.

3. Exam-Focused Summary

- **10-mark version:** Present the 7-step hypothesis testing procedure (from H_0/H_1 to Conclusion). Clearly list the formula for the t-statistic, show the calculation of $\bar{d}$ and s_d, and state the final result: **t-calculated is 17.37; Reject H_0**.

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4. b) Sample Size Calculation (6 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 7: Sampling Methods and Distributions
- **Page numbers:** Estimated Pages 190–223 (Section on sample size determination) [2]
- **Where the topic appears:** Within the context of calculating sampling error and determining required sample size for statistical inference.[2]

2. Quick Explanation

- **Core Idea:** Determine the minimum sample size (n) needed to estimate the population mean fuel efficiency within a desired margin of error (E), given a known population standard deviation (σ) and confidence level.
- **Required Inputs:**
  - Confidence Level (95%) $\rightarrow Z_{value} = 1.96$
  - Standard Deviation (σ) = 3 km/kWh
  - Margin of Error (E) = 1 km/kWh
- **Formula:** $n = \left(\frac{Z \cdot \sigma}{E}\right)^2$
- **Calculation:** $n = \left(\frac{1.96 \times 3}{1}\right)^2 = (5.88)^2 \approx 34.57$

3. Exam-Focused Summary

- **5-mark version:** State the formula, list the inputs ($Z=1.96$, $\sigma=3$, $E=1$), and show the steps: $n = (1.96 \times 3 / 1)^2 = 34.57$. Conclude by rounding up: **The required sample size is 35 electric vehicles.**

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4. Question 5 (20 Marks): ANOVA and Correlation

5. a) One-Way ANOVA (10 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 4: Design and Analysis of Experiments
- **Page numbers:** Pages 71–145 (Specifically, the section on Analysis of Variance) [4]
- **Where the topic appears:** Detailing the principles of partitioning variance (Sum of Squares) to test for differences in mean effects.[6]

2. Quick Explanation (Focus on setup and formulas)

- **Core Idea:** ANOVA (Analysis of Variance) tests the null hypothesis that the means of k populations (fertilizers A, B, C) are equal, by comparing the variance *between* the treatment groups to the variance *within* the groups.

- **Hypotheses:** $H_0: \mu_A = \mu_B = \mu_C$; $H_1:$ At least one mean is different.
- **Data Summary:**
  - A: 50, 52, 48, 51 ($\bar{X}_A = 50.25$)
  - B: 55, 54, 56, 57 ($\bar{X}_B = 55.50$)
  - C: 49, 47, 50, 48 ($\bar{X}_C = 48.50$)
  - Grand Mean ($\bar{\bar{X}}$) ≈ 51.42
- **Key Formulas (Required to be shown):**
  1. **Sum of Squares Total:** $SST = \sum_{i} \sum_{j} (X_{ij} - \bar{\bar{X}})^2$
  2. **Sum of Squares Treatment (Between Groups):** $SSTr = \sum_{i} n_i (\bar{X}_i - \bar{\bar{X}})^2$
  3. **Sum of Squares Error (Within Groups):** $SSE = SST - SSTr$
  4. **Mean Squares Treatment:** $MSTr = SSTr / (k-1)$
  5. **Mean Squares Error:** $MSE = SSE / (N-k)$
  6. **F-ratio:** $F = MSTr / MSE$
- **Expected Results (Conceptual):** Fertilizer B's yield (55.50) is notably higher than A (50.25) and C (48.50). This large difference between treatment means will likely result in a high F-ratio.

3. Exam-Focused Summary

- **10-mark version:** State the purpose and hypotheses. Set up the three columns of data. Present the required ANOVA table template (Source of Variation, SS, df, MS, F-ratio). Explicitly list the formulas for SST, $SSTr$, and the F-ratio.

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5. b) Pearson Correlation (10 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 10: Basic Multivariate Analysis
- **Page numbers:** Pages 366–419 (Specifically, the section on correlation and regression analysis) [4]
- **Where the topic appears:** Dealing with the measurement of association between two variables (Hours of Study and Scores).[4]

2. Quick Explanation

- **Core Idea:** Pearson's correlation coefficient (r) measures the strength and direction of the linear relationship between two quantitative variables (X: Hours, Y: Scores).
- Formula for r:

$$r = \frac{n \sum XY - (\sum X)(\sum Y)}{\sqrt{[n \sum X^2 - (\sum X)^2]}}$$

(Where $n=10$)

- **Calculation (Summary of values required):**
  - $\sum X = 59$

- $\sum Y = 663$
- $\sum X^2 = 413$
- $\sum Y^2 = 45293$
- $\sum XY = 4153$
- **Result:** The calculation yields a high positive correlation, $r \approx 0.957$.

- **Interpretation:** Since r is close to $+1$, there is a **strong positive linear relationship** between hours of study and exam scores. This means that as the number of hours studied increases, the exam score tends to increase significantly.

3. Exam-Focused Summary

- **10-mark version:** State the definition and range of r (from -1 to +1). Present the formula clearly. List the required summation values ($\sum X, \sum Y, \sum XY, \sum X^2, \sum Y^2$). State the calculated result ($r \approx 0.957$) and provide the necessary interpretation: "Strong positive relationship indicates increased study hours are highly predictive of increased scores."

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5. Question 7 (20 Marks): Chi-Square and Poisson Distribution

7. a) Chi-Square Test for Independence (14 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 8: Tests of Hypotheses
- **Page numbers:** Pages 224–310 (Specifically, the section on χ^2 tests) [4]
- **Where the topic appears:** Covering the use of χ^2 for testing independence of attributes (vaccination status and smallpox attack) and goodness of fit.[4]

2. Quick Explanation (Focus on setup and calculation steps)

- **Core Idea:** The Chi-square (χ^2) test determines if there is an association (dependence) between two categorical variables (Vaccinated/Not Vaccinated and Attacked/Not Attacked).
- **Hypotheses:**
  - H_0: Vaccination is **not** effective (Vaccination status and Attack status are independent).
  - H_1: Vaccination is effective (The statuses are dependent).
- **Contingency Table (Observed Frequencies, O_{ij}):**

Status	Attacked	Not Attacked	Total
Vaccinated	31	469	500
Not Vaccinated	185	1315	1500
Total	216	1784	2000

- **Calculation Steps (Required):**

1. Calculate **Expected Frequencies (E_{ij})** for each cell using $E_{ij} = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$.
    - $E_{11} (\text{Vax, Attacked}) = (500 \times 216) / 2000 = 54$
    - $E_{12} (\text{Vax, Not Attacked}) = (500 \times 1784) / 2000 = 446$
    - $E_{21} (\text{Not Vax, Attacked}) = (1500 \times 216) / 2000 = 162$
    - $E_{22} (\text{Not Vax, Not Attacked}) = (1500 \times 1784) / 2000 = 1338$
2. Calculate χ^2 statistic using the formula:

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

3. **Result (Conceptual):** The large difference between $O_{11}=31$ (Vaccinated, Attacked) and $E_{11}=54$ suggests a significant result. The calculated χ^2 value is approximately 25.84.
4. **Decision:** With $df=(r-1)(c-1) = 1$ and 5% significance, the critical χ^2 value is 3.841. Since $\chi^2_{calculated} (25.84) > \chi^2_{critical} (3.841)$, we **reject H_0**.
5. **Conclusion:** Vaccination is significantly effective.

3. Exam-Focused Summary

- **10-mark version:** Clearly state H_0 and H_1. Present the Observed Frequency Table and the Calculated Expected Frequency Table. State the χ^2 formula and the final outcome: $\chi^2_{calculated}$ is 25.84 (approx.). Reject H_0.

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7. b) Poisson Distribution Probability (6 Marks)

1. Location Finder

- **Book Name:** Panneerselvam (Research Methodology)
- **Chapter / Unit:** Chapter 6: Probability Distributions
- **Page numbers:** Pages 168–189 (Specifically, the section on discrete probability distributions) [4]
- **Where the topic appears:** Describing the parameters and function of the Poisson distribution for modeling the number of rare events (defects) in a fixed interval (per day).[4]

2. Quick Explanation

- **Core Idea:** The Poisson distribution models the probability of a given number of events (k) occurring in a fixed interval, provided these events occur with a known average rate (λ) independently of the time since the last event.
- **Inputs:**
    - Average rate (λ) = 2 defects/day
    - Desired number of events (k) = exactly 3 defects
- **Formula:** $P(X=k) = \frac{e^{-\lambda} \lambda^k}{k!}$
- **Calculation:**
    - $P(X=3) = \frac{e^{-2} \cdot 2^3}{3!} = \frac{0.1353 \times 8}{6} \approx 0.1804$

3. Exam-Focused Summary

- **5-mark version:** Define the Poisson distribution (modeling rare events). List the input parameters ($\lambda=2, k=3$). Present the formula for $P(X=k)$. State the result: **The probability of exactly 3 defective tires is approximately 0.1804 (or 18.04%).**